Image Super-Resolution Via a Convolutional Neural Network

CS229 Machine Learning

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Motivation and Overview

Image super-resolution attempts to counteract the effects of faulty imaging hardware or software degradation by interpolating between pixels. A super-resolution convolutional neural network (SRCNN) uses a pair of convolutional layers--a feature extraction layer and a feature reconstruction layer--to relate patches of low-resolution pixels to higher-resolution improvements.

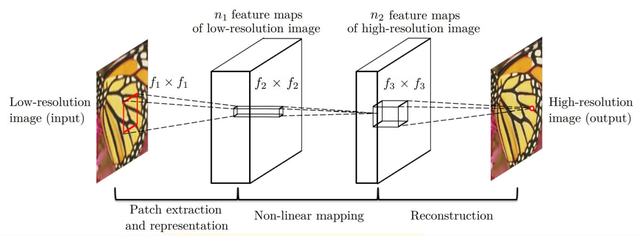


Figure 1. SRCNN architecture

The SRCNN which we are using was proposed by Dong et al. [1] and is a 3 hidden layer deep neural network. Patch extraction uses n_1=64 x 9x9 kernels, while reconstruction uses n_2=32 x 5x5 kernels.

Dataset: DIV2K

- Super-resolution specific dataset of large photographs (typical size 2040 x 1400 px)
- 100 photos: 60 training, 20 validation, 20 test

Preprocessing

- Crop to 800x800 for uniformity, resize to 224 x 224 px
- For inputs: downsize by 0.5 using bilinear interpolation, upsample by 2 for an input image of 224 x 224 px

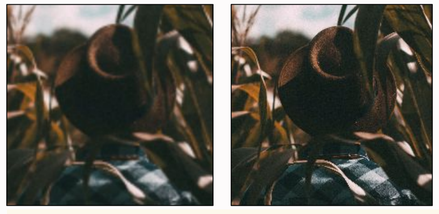


Figure 2. Example of downsampled input image (left) and target image (right)

Experiments

- Metric: peak signal-to-noise ratio $PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$
- Baseline: nearest neighbor, bilinear, bicubic upsampling
  - bicubic achieves highest test PSNR: 23.23
- Learning rate of 0.001 was most effective
  - highest test PSNR: 26.44
- `Imgaug` tensor transformations automatically normalized outputs, so tanh activation was not necessary
- Batch normalization was found not to improve SRCNN performance
- Regularization did not improve model performance

Number of Epochs vs. PSNR for Hyperparameters

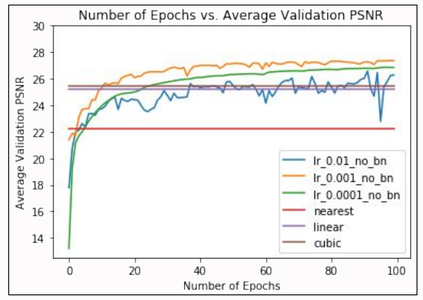


Figure 3. Basic SRCNN exceeds all of the baselines, with a learning rate of 0.001 achieving the best performance.

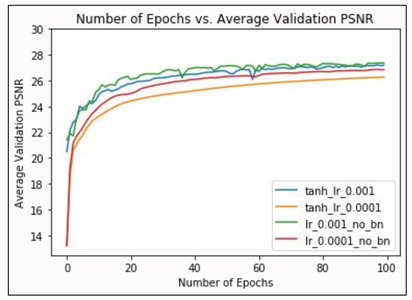


Figure 4. Tanh activation on the output did not improve validation PSNR.

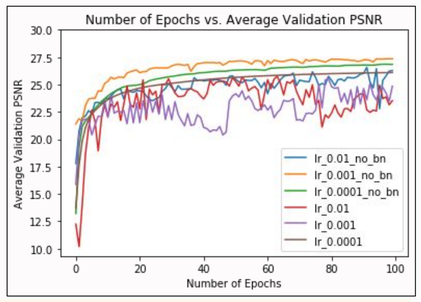


Figure 5. Batch normalization failed to improve PSNR.

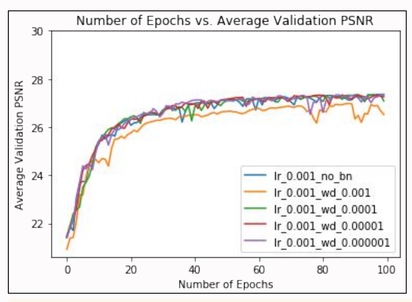


Figure 6. Regularization did not improve PSNR.

Model Summary

Model	Max Val PSNR	Test Loss	Test PSNR
Basic SRCNN	27.455	**0.0064324**	**26.442**
+Regularization	**27.459**	0.0064412	26.429
+Tanh	27.336	0.0066215	26.282
+BatchNorm	25.53	0.0100244	24.373
Bicubic	25.433	0.0066051	23.226
Bilinear	25.22	0.0074545	23.182
Nearest Neighbor	22.198	0.014747	20.551

Results



Figure 7: SRCNN inputs (left), predictions (center), targets (right)

SRCNN sharpens edges and adds naturalistic detail

- Some pixels still go to extreme colors, possibly due to small size of training dataset
- Some pixelation still evident in predictions

Future Work

- Test SRCNN with larger dataset (full DIV2K with 800 training images)
- Test response with noise added to input images
- With GPU availability, test U-Net and ResNet model ability to add more detail

References

[1] Dong, C., Loy, C.C., He, K. and Tang, X., 2015. Image super-resolution using deep convolutional networks. *IEEE transactions on pattern analysis and machine intelligence*, 38(2), pp.295-307.